

What the two counterexamples share

Exact diagonal-scaling comparison • 5 October 2026 • quarantined audit

Result: PASS, with a precise limitation

Both unnormalized overlap matrices are related by positive diagonal scaling. Main and independent exact checks agree. This establishes the shared algebraic pattern; it does not establish a geometric embedding or physical connection.

The general identity

For symmetric H and J with positive diagonals, define $N(H)_{ij} = H_{ij} / \sqrt{(H_{ii} H_{jj})}$. Then $N(H) = N(J)$ exactly when $J = S H S$, with $S_{ii} = \sqrt{(J_{ii} / H_{ii})} > 0$. Rearranging the normalized equality proves the forward direction; cancellation proves the reverse.

Consequently diagonal congruence is an algebraic restatement of equal normalized data. A pass here is not independent evidence of a deeper connection. The nontrivial additional fact in each original witness is that both overlap matrices arise from valid, nonisomorphic orders in their respective models.

Finite-poset witness: 16 labels

$H = Z_P Z_P^T$, $J = Z_Q Z_Q^T$. On either Boolean sector, $e_i = 2^{3-|X_i|}$ and $S_{ii} = \sqrt{8/e_i}$. All 256 entries satisfy $J = S H S$. An independent program reconstructs futures from subset rules instead of reading the zeta matrices.

Continuum witness: two events

$$H_P = \begin{bmatrix} 1/8 & 1/32 \\ 1/32 & 1/32 \end{bmatrix}.$$

$$H_Q = \begin{bmatrix} 1/16 & 1/32 \\ 1/32 & 1/16 \end{bmatrix}.$$

$$S = \text{diag}(1/\sqrt{2}, \sqrt{2}), \text{ so } H_Q = S H_P S.$$

Both normalize to $\begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix}$. The offdiagonal stays fixed because the scales multiply to one; the diagonals change.

Evidence, scope and sharing

What the second counterexample says

Two two-event samples in the same fixed open 1+1 Minkowski diamond have equal normalized overlaps of their full continuum future regions. One sample is a chain; the other is an antichain. The samples are therefore nonisomorphic as induced orders.

This uses continuum volumes, not counts of successors among the two sampled events. Individually attached future volumes distinguish the samples and restore the stated causal order. The full-domain E10 result uses all interior events and is not contradicted.

Reproduction and provenance

The predeclared plan, two executed Python scripts, exact matrices, output logs and run ledger are in AUDIT/. The source sharing ZIP is included unchanged in SOURCE/. Verification uses only the Python standard library and rational arithmetic; there was no floating-point approximation, variant search or solver.

Unzip and run `python reproduce_comparison.py`. New output goes to LOCAL_REPRODUCTION and is compared with both retained JSON certificates. Run `python verify_manifest.py` to check every packaged file hash. These are reader reruns, separate from the original audit executions.

Firewalls

Audit10 v144 and E11 remain frozen. The original sharing packet is preserved byte-for-byte. This comparison is not a canonical import or an E12 checkpoint. Any transfer requires a separate scoped audit of hypotheses, observation maps, proof and destination.

What remains open

No realization of the 16-element pair by continuum futures has been demonstrated here. No literature novelty claim or black-hole mechanism follows. A geometric realization question would be a separate next target, with its geometry fixed before testing; no such test was run.

For Saturday's planned post, the strongest supported lead is the explicit chain-versus-antichain continuum counterexample itself. The scaling comparison is explanatory context, not a claim of a new physical discovery.

Invariant and recovery map

Read this table as a selected certificate inventory, not a classification of everything recoverable from normalized data.

Quantity	Finite-poset pair	Continuum-sample pair
Cardinality	16 versus 16	2 versus 2
Normalized overlaps	All 256 G entries agree	Both $K=[1,1/2],[1/2,1]$
Strict comparable pairs	65 versus 65 (agrees)	1 versus 0 (differs)
Nonisomorphism certificate	Downset sizes among upset-size-two elements: {5,8,9,10} versus {6,7,8,11}	Chain versus antichain; comparable-pair count differs
Sizes / volumes	Upset-size multisets agree; attached vectors differ	Future-volume multisets: {1/8,1/32} versus {1/16,1/16}
Comparability degrees	Multisets differ; exact lists below	Induced sampled-order degrees: {1,1} versus {0,0}
Attached-data recovery	G plus individually attached upset sizes d: $x \leq y \text{ iff } Gxy^2 \text{ dx=dy}$	K plus individually attached future volumes a: $x \leq y \text{ iff } Kxy^2 \text{ ax=ay}$

Exact finite-poset degree multisets

P: [5,5,5,6,6,6,8,8,8,8,8,8,9,10,15,15]

Q: [5,6,6,6,6,6,7,7,8,8,8,8,11,15,15]

Order-isomorphism invariants must be distinguished from geometric data: future volumes and squared spacetime separations depend on the fixed continuum model. Equal normalized data do not imply equal orders. Diagonal scaling is equivalent to normalized equality and supplies no independent embedding claim.

Packet map: pages 1-2 comparison; page 3 this map; pages 4-5 finite-poset construction; pages 6-7 continuum proof. Each numbered counterexample folder has its own reader guide, input, certificates, original evidence and independent run instructions.

A counterexample to normalized poset reconstruction

Standalone research note • David Bridges • 1 October 2026

Result. Two nonisomorphic posets, each with 16 elements, have exactly the same labeled normalized principal-upset intersection matrix. This note records the construction and proof. It makes no novelty, minimality or publication claim.

1. What is being compared?

For a finite poset P , its principal upset is $U(x)=\{y : x\leq y\}$, including x . Put $K(x,y)=|U(x)\cap U(y)|$ and $d(x)=|U(x)|$. The normalized matrix is $G(x,y)=K(x,y)/\sqrt{(d(x)d(y))}$. A genuine collision means identical normalized data from posets that cannot be made identical by relabeling.

2. The attachment map

Take two disjoint copies E and W of the subsets of $\{0,1,2\}$. Each copy has eight labels. Define the map ϕ from E to W by the table below.

Subset X	$\phi(X)$
$\emptyset$	$\emptyset$
$\{0\}, \{1\}, \{2\}$	$\{0\}$
$\{0,1\}, \{0,2\}$	$\{0,1\}$
$\{1,2\}$	$\{0,2\}$
$\{0,1,2\}$	$\{0,1,2\}$

ϕ preserves inclusion and subset cardinality. Every singleton maps to $\{0\}$, and each pair maps to a pair containing 0. Write $e(X)=2^{(3-|X|)}$, the upset size inside a Boolean sector. Then $e(\phi(X))=e(X)$.

3. Define the two orders

Source P : use inclusion within both sectors, and $E_X\leq W_Y$ exactly when $\phi(X)\subseteq Y$. Candidate Q : use reverse inclusion within both sectors, and $E_X\leq W_Y$ exactly when $Y\subseteq\phi(X)$. Neither order has relations from W back to E .

Both are posets. Reflexivity and antisymmetry follow from the sector orders and one-way cross relations. Transitivity across sectors follows from inclusion preservation of ϕ . These are complete standalone examples; no embedding into a larger construction is assumed.

4. Exact equality of normalized matrices

For subsets A, B , let $k(A, B) = 2^{3 - |A \cup B|}$ and $\ell(A, B) = 2^{|A \cap B|}$. They count common successors and common predecessors in a Boolean sector, respectively. The cardinality identity $|A \cup B| + |A \cap B| = |A| + |B|$ gives:

$$e(A)e(B)\ell(A, B) = 8k(A, B).$$

Label pair	Source intersection K_P	Candidate intersection K_Q
E_X, E_Y	$k(X, Y) + k(\phi(X), \phi(Y))$	$\ell(X, Y) + \ell(\phi(X), \phi(Y))$
E_X, W_Y	$k(\phi(X), Y)$	$\ell(\phi(X), Y)$
W_X, W_Y	$k(X, Y)$	$\ell(X, Y)$

For a label i in either sector, let e_i be its sector upset size. Since ϕ preserves these sizes, every entry obeys $K_Q(i, j) = 8K_P(i, j)/(e_i e_j)$. Set $S(i, i) = \sqrt{8/e_i}$. Thus $K_Q = S K_P S$, including the diagonals.

The full diagonals are $d_P(E_X) = 2e(X)$, $d_P(W_X) = e(X)$, $d_Q(E_X) = 16/e(X)$, and $d_Q(W_X) = 8/e(X)$. These are the full-poset sizes used in normalization; $e(X)$ alone is a sector size.

Consequently $d_Q(i) = S(i, i)^2 d_P(i)$, and normalizing cancels both positive scale factors: $G_Q(i, j) = G_P(i, j)$ for every pair of labels.

5. The posets are not isomorphic

An order isomorphism preserves every element's upset and downset sizes. For the four elements with upset size two, the downset-size multisets are: $P: \{5, 8, 9, 10\}$; $Q: \{6, 7, 8, 11\}$. Therefore no relabeling can identify the orders.

In particular, $W_{\{0, 2\}}$ in P has two successors and nine predecessors: four from W and five from E . Q has no element with that pair of sizes. This is a complete nonisomorphism certificate.

6. Verification and scope

The frozen v126 construction and v127 audit include integer-matrix and independent set-based checks of all 256 entries, followed by cover-closure reconstruction and exact rational checks. An initial certificate label $W_{\{0, 1\}}$ was corrected to $W_{\{0, 2\}}$; the failed assertion and correction remain in the historical record.

Consequence: the universal claim that G determines every finite poset up to isomorphism is false. Earlier conditional results retain their hypotheses. The smallest collision size and literature novelty are unresolved. This document is an internal standalone note, not a preprint.

E11: Finite-sample overlap collision

Result: exact counterexample

Two two-event samples in the SAME fixed open 1+1 Minkowski diamond have identical continuum normalized future overlaps K but non-isomorphic induced causal orders. Thus K alone does not generally recover order from finite samples in this geometry. The full-interior E10 result remains valid.

Observation map and fixed geometry

Use $R=(0,1)$ times $(0,1)$ in null coordinates (u,v) , metric $ds^2=-du dv$ and volume measure $du dv/2$. K uses full continuum causal futures intersected with R . It does not count events in the sample. For an interior event x , $a_x=(1-u_x)(1-v_x)/2$; $c_{xy}=(1-\max(u_x,u_y))(1-\max(v_x,v_y))/2$; $K_{xy}=c_{xy}/\sqrt{a_x a_y}$.

The two samples

$P_1=(1/2,1/2)$, $P_2=(3/4,3/4)$. Both coordinates increase, so P_1 strictly precedes P_2 .
 $Q_1=(1/2,3/4)$, $Q_2=(3/4,1/2)$. One coordinate increases and the other decreases, so Q_1 and Q_2 are incomparable. All four events are strictly inside the same diamond.

Exact overlap certificate

P future volumes: $a_1=1/8$, $a_2=1/32$. Q future volumes: $b_1=b_2=1/16$. In both samples the offdiagonal intersection volume is $1/32$. Thus $K_P(1,2)=(1/32)/\sqrt{(1/8)(1/32)}=1/2$, and $K_Q(1,2)=(1/32)/\sqrt{(1/16)(1/16)}=1/2$. Both diagonals are 1. Both matrices are $[[1,1/2],[1/2,1]]$. Positive square roots are fixed by nonnegative overlap volumes.

Non-isomorphism and minimal size

The induced poset on P has one strict comparable pair; Q has zero. No relabeling can change that count, so the orders are not isomorphic. A one-event induced order has only one isomorphism type. Consequently two is the smallest possible sample size for this type of causal-order ambiguity. This minimality concerns sampled induced orders, not the size of Audit 10 poset collisions.

Checks, scope and what survives

Independent geometric check

In (t,z) , with $t=(u+v)/2$ and $z=(u-v)/2$, P is $(1/2,0),(3/4,0)$: squared interval $-1/16$. Q is $(5/8,-1/8),(5/8,1/8)$: squared interval $+1/16$. These establish timelike order versus spacelike separation directly. A separate rational verifier computes overlap ratios from future-side lengths and agrees with the main rectangular-future calculation.

Why E10 is not contradicted

Equal K on two stored query events need not extend to a K-preserving bijection of the entire diamond. E10 classified surjective full-domain isometries, and this two-point correspondence does not satisfy its hypotheses. Complete coverage and finite sampling therefore have different recovery guarantees.

Why E9 survives

The attached future-volume data differ: P has $1/8$ and $1/32$, Q has $1/16$ and $1/16$. With those individually attached volumes, E9 recovers each actual order through $K_{xy}^2 a_x = a_y$. This witness does not refute recovery with attached volumes, and does not address an unordered-volume-multiset question.

Executed provenance

The sample and criteria were recorded in PLAN_BEFORE_EXECUTION.md. Main check.py and independent_verify.py were each executed once with standard-library Fraction arithmetic. Both passed; no search, simulation, variant or rerun occurred. Original stdout/stderr, two exact JSON certificates and their field comparison are retained. Build_packet.py is executed artifact authoring, not an additional numerical experiment.

Physical and import limits

This is a geometric counterexample for exact continuum overlap data at finitely many query events. No hardware observation, detector model, black-hole mechanism or embedding of the original 16-element pair was asserted. E1-E10 evidence is preserved byte-identically and Audit 10 v144 remains untouched. E11 is completed pending freeze.

Reader map and next decision

Read the proof on page 1, then the exact scripts and results under E11/. Check the observation-map distinction before comparing with finite poset counts. No E12 is queued: after freeze, discuss whether to study recovery guarantees for additional data or broaden geometry, with a new predeclared target. Do not run either as part of this certificate.